DAR Assignment 1

You are supposed to work in groups of two persons.
Although this description is in English, you are allowed to write your report in Dutch.

For assignment 1 you will have to establish a connection between a Python- or C#-program and a database (SQLite).

Ranking on query results

The goal of this assignment is basically to implement the ideas of the article by Agrawal, Chaudhuri e.a. *So you have to solve both the zero-answers and the many-answers problem.* We offer a table and a query workload. They are provided on Blackboard.

The program should be able to process conjunctive equality queries (ceq) on the table. A ceq consists of predicates of the kind *attr = value*, separated by comma's, terminated by a semicolon. An input consists of a ceq and is read by an interface. Keep this interface simple (command line suffices). The required value for *k* is also entered according to this syntax. When missing, use a default value *k* = 10. Example inputs:

k = 6, brand = 'volkswagen';
cylinders = 4, brand = 'ford';

The basic query is
*SELECT * FROM autompg WHERE ceq;*
In the ceq, the comma's are replaced by ANDs and the *k*-value is left out.
The output consists of the top-*k* tuples according to some ranking principle.

In fact, you have to write two programs. The first program will do some preprocessing on the data and/or the workload. During this phase, a metadatabase will be constructed and filled. This metadb will be used when answering the actual queries. The second program will be able to process queries and show the answers, making use of the metadb for ranking. Note that your metadb should be constructed only *once*, for this particular contents of the db and before processing a batch of queries.

Your software is supposed to meet at least requirement [1] below. Then your maximum score is 8. If you deal in a satisfying way with requirements [2] and/or [3], you can increase your maximum score with one point for each requirement. See the evaluation form on Blackboard for more detailed information. Please pay special attention to which attributes should be seen as numerical.

[1] Deal with similarity properties. Pay also attention to the numerical attributes. Demonstrate your findings.
[2] Use sophisticated techniques for finding value-similarities.
[3] Use sophisticated techniques for top-k calculations.

The **deliverables** are:
- A C#/Python program to determine and fill the contents of the metadb, based on the preprocessing of data and the workload.
- A C#/Python program to deal with the queries.

- A text file metadb.txt, containing the data definitions required for your meta database (that is filled during the preprocessing).
- A text file metaload.txt, containing sql-statements used to fill the metadb.
- A description of your approach, explaining choices and describing experiences. It contains a class diagram of your second program. It also contains an extensive discussion of your approach towards solving both problems. Format: pdf; max 6 pages. This report is also important when grading your work.
- Finally, you will have to give a short demonstration of your program to one of the student assistants. Make sure that either you or your group member is present. Deadline of the demo: see Schedule on Blackboard.

You can choose whether you want to write your program in C# or Python. For this assignment, there are templates available on Blackboard. One is in C#, and one is in Python. The templates are meant as a guide to get an idea on designing your assignment. You do not have to fill them in. **It is strongly recommended to write your own code!**

Zip your files before submitting the assignment under *Assignments* on Blackboard. It suffices if only one group member submits it on Blackboard. The zip file should clearly state the names of the group members. The feedback after grading will be put on Blackboard to the one who submitted it. Deadline: see Schedule on Blackboard.

The retake is the same assignment. However, the maximum grade is 7.0. It is recommended to use the received feedback from the first attempt to improve the assignment for the retake. If you have taken one or both lab exercises before and received a grade of 7.0 or higher, then you can get an exemption. Please contact the practicum coordinator for this.

Your code and report will be checked by the UU's plagiarism tool!